1 Totally Ordered Sets

Remind yourself of strict total order from Lecture 7.

Definition 1.1. A set equipped with a strict total order is called a totally ordered set.

Example 1.1. On $\mathbb{N}$ define a strict total order as

$$a < b \iff \exists n \in \mathbb{N} \quad \text{such that } a + n = b.$$

Then $(\mathbb{N}, <)$ is a totally ordered set.

Example 1.2. On $\mathbb{Z}$ define a strict total order as

$$a < b \iff \exists n \in \mathbb{N} \quad \text{such that } a + n = b.$$

Then $(\mathbb{Z}, <)$ is a totally ordered set.

Example 1.3. On $\mathbb{Q} = \left\{x = \frac{a}{b} : a \in \mathbb{Z},\ b \in \mathbb{N}\right\}$, define a strict total order as

$$\frac{a}{b} < \frac{c}{d} \iff ad < bc \ \text{ in } \mathbb{Z}.$$

$(\mathbb{Q}, +, \cdot, <)$ (with the usual addition $+$, the usual multiplication $\cdot$, and usual order $<$ defined as above) is an ordered field. Remind yourself of the axioms that an ordered field must satisfy from Lecture 7.

2 Completeness

We will now get to the last axiom we need to characterize the real numbers: *completeness*. To state this axiom, we first need to introduce some definitions. The definitions below apply generally to totally ordered sets, and in particular, they apply to ordered fields. Recall that we have defined $x \leq y$ to mean $x < y$ or $x = y$, and $x > y$ to mean $y < x$.

2.1 Upper and lower bounds; supremum and infimum

Definition 2.1. A subset S of a totally ordered set T is

- **upper bounded** if there exists $u \in T$ such that $s \leq u$ for all $s \in S$;
  Then u is an **upper bound** of S.
- **lower bounded** if there exists $l \in T$ such that $l \leq s$ for all $s \in S$;
  Then l is a **lower bound** of S.
- **bounded** if it is both upper bounded and lower bounded.

Remark 1. Let $S \subseteq T$, and let $u \in T$ be an upper bound of S; that is, $s \leq u$ for every $s \in S$. Every element u_1 of T with $u_1 \geq u$ is also an upper bound of S: if $s \in S$, then

$$s \leq u \leq u_1.$$

Similarly, if l is a lower bound of S, then every element l_1 of T with $l_1 \leq l$ is also a lower bound of S: if $s \in S$, then

$$l_1 \leq l \leq s.$$

Definition 2.2. Let S be a subset of a totally ordered set T.

(i) If S is upper bounded, we say that $u^* \in T$ is a **supremum** or **least upper bound** of S if

- u^* is an upper bound of S, and
- if u is another upper bound of S, then $u^* \leq u$.

If u^* exists, then we write

$$u^* = \sup(S).$$

(ii) If S is lower bounded, we say that $l^* \in T$ is an **infimum** or **greatest lower bound** of S if

- l^* is a lower bound of S, and
- if l is another lower bound of S, then $l \leq l^*$.

If l^* exists, then we write

$$l^* = \inf(S).$$

Proposition 2.1. Let S be a subset of a totally ordered set T. Suppose that S has a supremum $u^* = \sup(S)$. Then the supremum u^* is unique.

Proof. □

Example 2.1. Consider the subset $H_+ = \left\{\frac{1}{n} \in \mathbb{Q} : n \in \mathbb{N}\right\}$ of the set $\mathbb{Q}$ of rational numbers. Prove that $\sup H_+ = 1$ in two steps:

(i) Prove that 1 is an upper bound for H_+ in $\mathbb{Q}$.

(ii) Prove that if u is an upper bound for H_+ in $\mathbb{Q}$, then $1 \leq u$.

Remark 2. Here, the candidate 1 for the supremum happened to be available as an element $1 \in \mathbb{Q}$, and it is also an element of the subset itself: $1 \in H_+$.

Example 2.2. Consider the subset $H_- = \left\{-\frac{1}{n} \in \mathbb{Q} : n \in \mathbb{N}\right\}$ of the set $\mathbb{Q}$ of rational numbers. Prove that $\sup H_- = 0$ in two steps:

(i) Prove that 0 is an upper bound for H_- in $\mathbb{Q}$.

(ii) Prove that if u is an upper bound for H_- in $\mathbb{Q}$, then $0 \leq u$.

Remark 3. Here, the candidate 0 for the supremum happened to be available as an element $0 \in \mathbb{Q}$, but it is *not* an element of the subset itself: $0 \notin H_-$.

We have just seen that the supremum of a set S may or may not be an element of S. This leads to the following definition.

Definition 2.3. Let S be a subset of a totally ordered set T.

(i) If S has a supremum u^* and $u^* \in S$, then we call u^* the **maximum** of S and write

$$\max(S) = u^*.$$

(ii) If S has an infimum l^* and $l^* \in S$, then we call l^* the **minimum** of S and write

$$\min(S) = l^*.$$

Definition 2.4. Let $(T, <)$ be a totally ordered set. We say that T is *complete* if every nonempty subset $S \subseteq T$ that is upper bounded has a supremum in T.

Definition 2.5. Let $(F, \oplus, \otimes, <)$ be an ordered field. Then F is a *complete ordered field* if every nonempty subset $S \subseteq F$ that is upper bounded has a supremum in F.

Remark 4. Completeness is a property of a totally ordered set; it does not require a field structure. A complete ordered field is an ordered field whose underlying totally ordered set is complete.

Remark 5. Let T be a complete totally ordered set. Then every nonempty subset $S \subseteq T$ that is lower bounded has an infimum in T. Why?

2.2 The complete ordered field $\mathbb{R}$

Theorem 2.2 below shows that $(\mathbb{Q}, +, \cdot, <)$, with the usual addition, multiplication and order, is not a complete ordered field. We will first need to prove the following theorem.

Theorem 2.1. There is no $q \in \mathbb{Q}$ such that $q^2 = 2$.

Proof. Suppose that there exists $q \in \mathbb{Q}$ such that $q^2 = 2$. Replacing q by $-q$, if necessary, we may assume that $q > 0$ (note that $q \neq 0$ since $0^2 = 0 \neq 2$.) We can write q in its reduced form, i.e. in the form $q = \frac{m}{n}$, where $m, n \in \mathbb{N}$ and the only common divisor of m and n is 1. From $q^2 = 2$ we obtain

$$m^2 = 2n^2,$$

so m^2 is even, and thus m must be even (the square of an odd number is odd- prove this as a short exercise). This means that $m = 2k$ for some $k \in \mathbb{N}$. Then

$$n^2 = 2k^2,$$

so n^2 is even, and thus n must also be even. This is a contradiction: m and n both being even implies that they have 2 as a common divisor, while we had supposed that the only common divisor of m and n was 1. □

Remark 6. Notice that by the Pythagorean Theorem, the hypotenuse of a right triangle whose sides have length 1 has length x satisfying

$$x^2 = 1^2 + 1^2 = 2.$$

If every length could be measured by a rational number, then the length x would have to correspond to some rational number q. So the above theorem tells us that there are *lengths* that cannot be measured by rational numbers. If we want a number system that is sufficient for measuring any distance on the line, then we must *extend* the set of rational numbers.

Theorem 2.2. The field $\mathbb{Q}$ is **not** a complete ordered field: there exists a nonempty subset S of $\mathbb{Q}$ that is upper bounded but has no supremum (no least upper bound).

Proof. Consider the set

$$S = \{x \in \mathbb{Q} : x > 0 \quad \text{and } x^{\,2} < 2\}.$$

First observe that S is nonempty: we have $1 \in S$ (because $1 > 0$ and $1^2 = 1 < 2$).
Using the fact that if x, y are positive rational numbers, then $x < y$ if and only if $x^2 < y^2$, we have the following observation:
Observation: if $y^2 > 2$ for $y > 0$, $y \in \mathbb{Q}$, then y is an upper bound of S.
It follows that S is upper bounded: for example, 3 is an upper bound of S, since $3 > 0$,$3 \in \mathbb{Q}$ and $3^2 = 9 > 2$.
We suppose that S has a least upper bound $u^* \in \mathbb{Q}$, and we will arrive at a contradiction.
Since $1 \in S$ and u^* is an upper bound of S, we have $u^* \geq 1 > 0$.
According to Theorem 2.1 there is no rational number whose square is equal to 2. So since $u^* \in \mathbb{Q}$, $(u^*)^2 = 2$ is impossible. By the trichotomy axiom of order, necessarily one of $(u^*)^2 > 2$ or $(u^*)^2 < 2$ must hold.

(i) Suppose that $(u^*)^2 > 2$. We will find $0 < \varepsilon < u^*$ such that

$$(u^* - \varepsilon)^2 > 2.$$

Then by the observation, $u^* - \varepsilon$ will be an upper bound of S, and at the same time $u^* - \varepsilon < u^*$. Then there exists an upper bound that is strictly smaller than the least upper bound, a contradiction.

Choice of ε: We want $0 < \varepsilon < u^*$ and

$$(u^* - \varepsilon)^2 = (u^*)^2 - 2u^*\varepsilon + \varepsilon^2 > 2.$$

Since $\varepsilon^2 > 0$, it is enough to ensure that

$$(u^*)^2 - 2u^*\varepsilon > 2,$$

which is equivalent to

$$\varepsilon < \frac{(u^*)^2 - 2}{2u^*}.$$

Observe that $((u^*)^2 - 2)/(2u^*)$ is a positive rational number. Thus, if we choose

$$\varepsilon = \frac{1}{2} \min \left\{ u^*, \frac{(u^*)^2 - 2}{2u^*} \right\},$$

then we have found a rational ε satisfying $0 < \varepsilon < u^*$ and $(u^* - \varepsilon)^2 > 2$.

(ii) Suppose that $(u^*)^2 < 2$. We will find a rational $\varepsilon > 0$ such that

$$(u^* + \varepsilon)^2 < 2.$$

Then $u^* + \varepsilon > u^*$ and $u^* + \varepsilon \in S$. So there exists an element in S that is strictly larger than the supremum of S, a contradiction.

Choice of ε: We want $\varepsilon > 0$ and

$$(u^* + \varepsilon)^2 = (u^*)^2 + 2u^*\varepsilon + \varepsilon^2 < 2.$$

We choose $\varepsilon \leq 1$, in which case

$$(u^*)^2 + 2u^*\varepsilon + \varepsilon^2 \leq (u^*)^2 + 2u^*\varepsilon + \varepsilon = (u^*)^2 + \varepsilon(2u^* + 1),$$

because $\varepsilon^2 \leq \varepsilon$. It is therefore enough to ensure that

$$(u^*)^2 + \varepsilon(2u^* + 1) < 2,$$

which is equivalent to

$$\varepsilon < \frac{2 - (u^*)^2}{2u^* + 1}.$$

Observe that $(2 - (u^*)^2)/(2u^* + 1)$ is a positive rational number. Thus, if we choose

$$\varepsilon = \frac{1}{2} \min \left\{ 1, \frac{2 - (u^*)^2}{2u^* + 1} \right\},$$

then we have found a rational $\varepsilon > 0$ satisfying $(u^* + \varepsilon)^2 < 2$.

Assuming that S has a supremum $u^* \in \mathbb{Q}$, we have ruled out $(u^*)^2 = 2$, $(u^*)^2 > 2$, and $(u^*)^2 < 2$. Therefore S has no supremum (in $\mathbb{Q}$).

□

All the work we will do in this course is based on the following extension theorem.

Theorem 2.3. There exists a unique (up to isomorphism) complete ordered field, $(\mathbb{R}, +, \cdot, <)$. $(\mathbb{R}, +, \cdot, <)$ contains a subfield that is isomorphic to $(\mathbb{Q}, +, \cdot, <)$.

That is, there exists a unique (up to isomorphism) complete ordered field $(\mathbb{R}, +, \cdot, <)$, and it contains (an isomorphic copy of) the rational numbers. The set $\mathbb{R}$ is the set of **real numbers**. The operations $+$ and $\cdot$ on $\mathbb{R}$ extend the corresponding operations on $\mathbb{Q}$ and satisfy the axioms of addition (A1,A2,A3,A4) , the axioms of multiplication (M1,M2,M3,M4), and the distributive law (D). The order $<$ on $\mathbb{R}$ extends the order on $\mathbb{Q}$ and satisfies the order axioms (O1,O2, O3, O4). Moreover, $\mathbb{R}$ is a complete ordered field, i.e. it satisfies the completeness axiom.
Completeness axiom for the real numbers. Every nonempty subset S of $\mathbb{R}$ that is upper bounded has a supremum.

Remark 7. Uniqueness up to isomorphism in Theorem 2.3 means that if $(F_1, \oplus, \otimes, <)$ is another complete ordered field, then there exists a bijection

$$g : F_1 \longrightarrow \mathbb{R}$$

such that for all $a, b \in F_1$,

(i) $g(a \oplus b) = g(a) + g(b)$;

(ii) $g(a \otimes b) = g(a)g(b)$;

(iii) if $a > b$, then $g(a) > g(b)$.

This means that any complete ordered field F_1 is essentially a copy of the complete ordered field $(\mathbb{R}, +, \cdot, <)$.

We will accept Theorem 2.3 for now and possibly return to it later. Check the supplemental lecture notes for more related to this extension.
It is now possible to show the following theorem.

Theorem 2.4. There exists a unique positive $x \in \mathbb{R}$ such that $x^2 = 2$.

Remark 8. Theorem 2.4 gives us the existence of a real number x such that $x^2 = 2$, i.e. the existence of the root of 2 in the set of real numbers. You can try to prove it, hint: use the completeness axiom. Or, you can read the proof from the supplemental lecture notes. By Theorem 2.1, this real number x such that $x^2 = 2$ is not a rational number. Consequently, there exist real numbers that are not rational. These are called *irrational numbers*. We denote the set of irrational numbers by $\mathbb{R} \setminus \mathbb{Q}$.